Statistical Numeracy and Belief Bias: A Case of Medical Decision Making

Clayton Ulm[1,2], Jinhyo Cho,[1,3] Edward T. Cokely,[1, 3, 4]

[1] Department of Psychology, University of Oklahoma, USA.
[2] TSET Health Promotion Research Center, University of Oklahoma Health Sciences Center, USA.
[3] National Institute for Risk & Resilience, University of Oklahoma, USA.
[4] Max Planck Institute for Human Development, Germany.

DECISION ANALYTICS LAB
www.DecisionAnalyticsLab.org
claytonulm@ou.edu

Belief Bias in Medical Decision-Making

- Use of prior beliefs can increase efficiency via increased speed and reduced deliberation, but often leads to ignoring medical data and relying instead on prior beliefs (i.e., **belief bias**) to influence choice, which may detract from decision quality.
- Belief bias is often associated with less normative decisions due to base-rate neglect, worsened metacognitive ability, and low conflict detection.

Numeracy in Medical Decision-Making

- 29% of American adults cannot perform beyond single-step mathematic processes like counting and understanding percentages like 50%.
- In medical decision-making, low numeracy predicts reduced ability to understand medical problems, interpret data and apply it to personal context, but also increased tendency to ignore data.
- Prior research has captured belief bias in think-aloud protocols, yet few studies have examined if low numeracy increases vulnerability to belief bias.

Does numeracy predict vulnerability to belief bias in medical decision-making?

Methods

- Undergraduates from the University of Oklahoma ($N = 107$)
  - Berlin Numeracy Test-Schwartz to assess numeracy
  - One Medical problem that beckons simple conversion from frequency to percentage to assess decision quality
  - Overconfidence Rating ($Confidence\ Rating - 10 * Choice\ Accuracy$)
  - Write-Aloud Protocol (see coding table below)

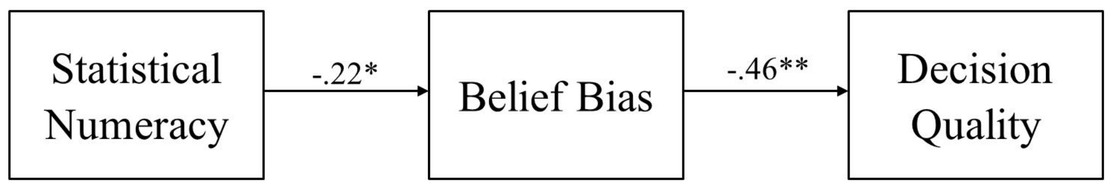
Belief Bias	Coded Reasoning Example
Present	"I just assumed the cancer pill more effective because people had higher chances of dying to cancer"
Absent	"28/1000 equals a 2.8% chance of dying from cancer. 59/1000 equals a 5.9% chance of dying from a stroke. The chance of dying from a stroke is greater than for cancer. Therefore, the person should take medication against stroke over cancer."

- Two independent raters coded reasonings and inter-rater reliability was high, $r(80) = 0.93, p < 0.001$

Relationships Between Variables

- Those who engaged in belief bias tended to have lower decision quality, $r(105) = -0.46, p < 0.001$
- Those who engaged in belief bias tended to have lower numeracy scores, $r(105) = -0.22, p = 0.043$
- No relationship was found between numeracy and decision quality, $r(105) = 0.12, p = .24$

Numeracy Indirectly Predicts Decision Quality

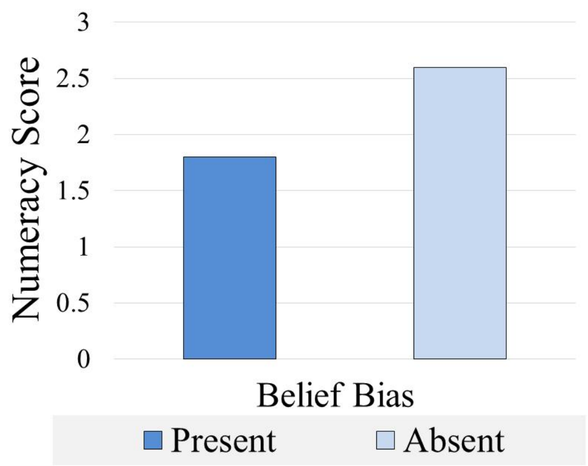


$\chi 2(2) = .56$, $p = .76$, CFI = 1.00, TLI = 1.00, SRMR = .00, RMSEA = .00 with 90% C.I (0.00-0.08)

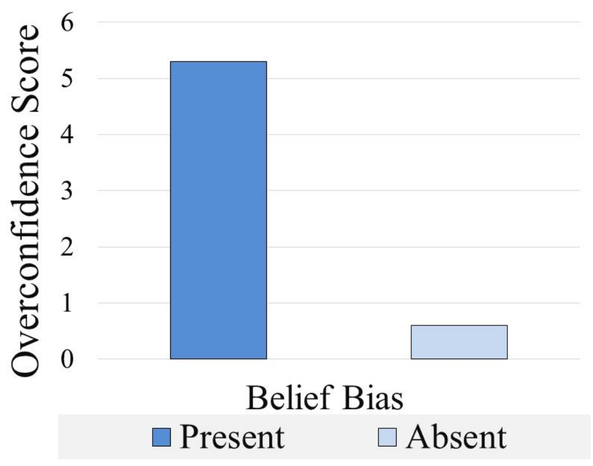
Reported are standardized coefficients

- 5000 bootstrap sample using lavaan package in R Studio 1.1.383
- Structural model analysis confirmed an **indirect effect** of numeracy on decision quality through presence of belief bias, .10, $p < .05$ with 90% CI (002 - .07)

Comparing Belief Bias, Numeracy and Overconfidence



- Those who engaged in belief bias tended to have lower numeracy scores
- $t(22) = 2.05$, $p = 0.046$, $d = 0.55$



- Those who engaged in belief bias tended to be more overconfidence than those who did not
- $t(83) = -3.57$, $p < 0.001$, $d = -0.97$

Discussion

- **Numeracy mitigates ignorance:** Numeracy tends to predict decision quality since higher numerate individuals tend not to utilize prior belief and ignore data when present.
- **Data is important:** Although employing belief biases may subvert excessive deliberation, when the data is incongruent to the bias, further deliberation is necessary to increase decision quality.
- **Future research:** More research should examine if vulnerability to belief bias is related to numeracy in financial decision-making.

References

Cokely, E. T., Feltz, A., Ghazal, S., Allan, J. N., Petrova, D., & Garcia- Retamero, R., (2018). Skilled Decision Theory: From intelligence to numeracy and expertise. In K. A. Ericsson, R. R. Hoffman, A. Kozbelt, & A. M. Williams (2nd Eds.), Cambridge Handbook of Expertise and Expert Performance. New York, NY: Cambridge University Press.

Cokely, E. T., Galesic, M., Schulz, E., Ghazal, S., & Garcia-Retamero, R. (2012). Measuring Risk Literacy: The Berlin Numeracy Test. *Judgment & Decision Making, 7*(1).

Cokely, E. T., & Kelley, C. M. (2009). Cognitive abilities and superior decision making under risk: A protocol analysis and process model evaluation. *Judgment and Decision Making*, *4*(1), 20–33.

De Neys, W., & Glumicic, T. (2008). Conflict monitoring in dual process theories of thinking. *Cognition*, *106*(3), 1248–1299. https://doi.org/10.1016/j.cognition.2007.06.002

De Neys, W., Cromheeke, S., & Osman, M. (2011). Biased but in Doubt: Conflict and Decision Confidence. *PLOS ONE*, *6*(1), e15954. https://doi.org/10.1371/journal.pone.0015954

Desjardins, R., Thorn, W., Schleicher, A., Quintini, G., Pellizzari, M., Kis, V., & Chung J., E. (2013). OECD Skills Outlook 2013: First Results From the Survey of Adult Skills. *Journal of Applied Econometrics*, *30*(7), 1144–1168.

Reyna, V. F., Nelson, W. L., Han, P. K., & Dieckmann, N. F. (2009). How Numeracy Influences Risk Comprehension and Medical Decision Making. *Psychological Bulletin*, *135*(6), 943–973. https://doi.org/10.1037/a0017327

NATIONAL INSTITUTE for RISK and RESILIENCE The UNIVERSITY of OKLAHOMA

MAX-PLANCK-GESELLSCHAFT

TSET Health Promotion Research Center

OU Health | Stephenson Cancer Center

RISKLITERACY.ORG Science for Informed Decision Making